

QI × Boston University

AI & Data-Engineering Ecosystem Fit Assessment

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Subject: BU AIDA / Atlas / Nexus environment vs. the QI toolset — focus MapSnap & CogniBase

Classification: Quiddity Innovations INTERNAL ONLY — not for external distribution

Rev 2 — adds a topology deep-dive from the Nexus page-3 diagram (visual detail not in the text layer).

0. Executive Summary

Boston University's IS&T organisation has stood up one of the most architecturally disciplined enterprise-AI programmes in higher education — the AIDA initiative, built on a single open-source, cloud-native platform (“Atlas”) that powers two product families: TerrierAI (community AI) and TerrierData (institutional data). The defining posture is a “critical embrace” of AI: architecture first, curated heterogeneity, no vendor lock-in, and a literal selection test of “strategic fit, cost, capability, and freedom to leave.” The entire stack runs at \$0/yr in software licensing (97% open source).

Assessed against Quiddity Innovations' current toolset, the headline finding is unusually strong: three QI products already point directly at BU's exact technology surface, and several of QI's platform choices line up with BU's by independent convergence rather than coincidence.

The three direct-fit products

QI product	Targets at BU	Why it fits
CogniBase	Hyland OnBase (BU runs it; BU has a REST POC team)	Brings the OnBase domain layer (DocType modelling, Custom Query semantics, Normalizers, Gates, Query Federator, three-corpus RAG) that BU's REST team explicitly lacks.
MapSnap	Jenzabar EX + OnBase + any SQL Server/Oracle/Postgres schema	BU is retiring Jenzabar and stabilising SIS — MapSnap turns an undocumented 2,000-table schema into plain-English answers and NL→SQL. A migration-archaeology tool.
AutoPDF	OnBase ingestion (emits OnBase XML Index DIP, Foundation 24.1)	Local-first batch convert / split / extract → records-team feeder for OnBase. Confidential docs never leave the machine — FERPA-safe by construction.

The convergence signals (QI and BU already chose the same things)

- OpenClaw — BU's TerrierAI Assistant is built on OpenClaw; Renne already runs OpenClaw in production (C:\OC). Renne holds operational OpenClaw experience BU is only now hardening.
- MCP (Model Context Protocol) — BU's Nexus/Regent agents call data through MCP tools; QI's Brain and tooling are already MCP-native. This is the single cleanest integration surface.
- Claude Code — BU uses a custom Claude Code skill to convert SnapLogic→Prefect pipelines; QI's entire operating model is Claude-Code-driven.
- Vector + open formats — BU uses pgvector / Iceberg / Trino; QI's tools use ChromaDB/pgvector-class stores and are designed engine-agnostic.

The one collision to fix

BU's AI platform is named “Nexus.” Renne also has a project named NEXUS (C:\NEXUS). In any BU-facing material this will cause confusion and should be renamed or clearly differentiated.

Bottom line

The strongest, lowest-friction wedge is CogniBase as the OnBase domain layer for BU's REST POC team, with MapSnap + AutoPDF bundled as a Jenzabar/OnBase migration toolkit. The price of entry is conformance to BU's architecture rules — open standards, MCP-native, K8s/ARM-deployable, Entra ID identity, audit-first, and no per-seat licensing. CogniBase already meets most of these by design; MapSnap and AutoPDF need a container/MCP packaging pass. Commercially, lead with a forward-deployed design partnership (the CollegeVine model) in the operational/IT-data niche CollegeVine does not touch, backed by flat-fee, no-lock-in tool licences — never per-seat.

1. Method & Inputs

This assessment cross-reads two BU source documents against the live QI ecosystem registry (25 registered projects) and the on-disk design/intro documentation for the BU-relevant products.

Sources analysed

Input	What it provided
BU-AI-Data-Engineering-Dossier-for-Quiddity.md	Full BU site crawl: AIDA strategy, four-tier portfolio, TerrierAI/TerrierData products, the eight Atlas components, full tech stack, 2026 roadmap (4 lanes), vendors being retired.
nexus-platform-overview.pdf (8 pp, BU IS&T, Apr 2026)	Deep technical spec of BU's Nexus AI-chat platform: 5-layer topology, MCP server, agent loop, admin portal, LLM providers, infra/security/observability.
C:\QIH\ecosystem\qi_registry.json	Authoritative list of 25 QI projects, ports, status, families.
C:\CogniBase\DESIGN* + README	CogniBase architecture, three-corpus model, access model, Query Federator, and the BU REST-POC coordination + beyond-OnBase strategy docs.
C:\MapSnap\INTRO + docs	MapSnap status, knowledge-bundle design, OnBase/Jenzabar targeting, BU video script.
C:\AutoPDF\INTRO	AutoPDF local-first PDF toolkit + OnBase DIP output.

2. The BU Target Environment (condensed)

The four-tier AI portfolio — where a partner can land

Tier	Name	Occupant / nature	Door for QI?
01	Bleeding edge	Frontier agentic design partnerships — today only CollegeVine (admissions/recruiting).	OPEN — in the operational / IT-data niche CollegeVine does not cover.
02	Homegrown	BU's own TerrierAI/TerrierData on open-source foundations.	Capability injection, not product sale.
03	Enterprise platform AI	AI inside Salesforce/SAP/ServiceNow/Blackboard.	Indirect — OnBase-adjacent.
04	Specialized applications	Niche vended tools with embedded AI, adopted case-by-case.	OPEN — natural home for MapSnap / AutoPDF / CogniBase as flat-fee tools.

The target “Intelligence Stack” (every agent reasons through these; never touches raw systems)

- Agent layer — reasons, plans, executes.
- Knowledge base — curated, domain-owned, one source of truth.
- Semantic layer — raw data → shared business terms.
- Ontology — the institutional map of how entities relate (BU's self-identified weakest, most strategic layer).

The eight Atlas components (and which QI tool relates)

Atlas component	Role / tech	QI relevance
Forge	K8s (EKS)+Graviton/ARM, ArgoCD, Linkerd, OPA	Deployment target — QI tools must be container/ARM-friendly.
Cortex	Iceberg-on-S3, Spark, Dagster, Trino, dbt (Data Lake)	CogniBase's “lift OnBase into a lake” mission lands here if it emits Iceberg/Trino-queryable output.
Lattice	APISIX gateway, FastAPI microservices	QI tools are FastAPI — same idiom; expose via APISIX.
Convoy	Prefect batch + Claude Code conversion skill	AutoPDF/MapSnap pipelines could run as Prefect flows.
Meridian	EventBridge, SNS/SQS, Person Broker (400+ attrs)	Event consumers; CogniBase Normalizers parallel the Person CDM.
Nexus	AI platform: FastAPI, MCP, multi-provider (Claude/GPT/Gemini)	Primary integration host — CogniBase/MapSnap as MCP tool endpoints Nexus agents call.
Lexicon	Cube, Superset, OpenMetadata (semantic layer)	Ontology/knowledge-graph work — QI semantic-modelling play.
Regent	Headless AI workflows on Nexus agent framework + MCP (PLANNED)	Least mature → largest open frontier for agentic IP.

The architecture rules = the price of entry

- Open-standard native — Iceberg tables, Trino-queryable, MCP tool interfaces, OPA-enforceable. No closed formats.
- No per-seat licensing; no lock-in — “freedom to leave” is a literal selection criterion.
- Identity-aware & auditable — Entra ID SSO, per-user isolation, audit logging (requirements, not nice-to-haves).
- Data-classification guardrails — no HIPAA/PCI in the lake; FERPA-grade auditing.
- Cloud-native, ARM-friendly, K8s-deployable — to align with Forge (EKS + Graviton).
- Forward-deployed delivery — lean embedded engineers over heavy vendor relationships.

Vendors BU is actively removing (budget is being freed)

SnapLogic (~\$450K/yr being eliminated), MicroStrategy (hosting migration / re-licensing), Jenzabar (retirement), Buzz (replacement), mainframe archives. A well-priced open tool that helps these migrations can capture a slice of freed budget — and avoids looking like the legacy middleware BU is shedding.

2A. Topology Deep-Dive — what the Nexus diagram reveals (Rev 2)

The Nexus overview's only embedded graphic — the page-3 platform-topology diagram — carries detail that the document's text never states. Reading it closely materially sharpens the integration picture and confirms several assumptions. The most important revelation is the physical two-VPC separation and the exact seam through which AI reaches data.

The diagram (BU Nexus platform topology, page 3)

MCP Server nexus-mcp-server	Exposes BU data as structured tools (search, lookup, submit) using the Model Context Protocol standard. Each department gets its own tool endpoint.	Automated (called by agent)
Data Layer pgvector, Trino, APIs	Vector databases, relational stores, and live APIs that contain BU content — course catalogs, WordPress sites, ServiceNow, student systems.	Automated (indexed by pipelines)

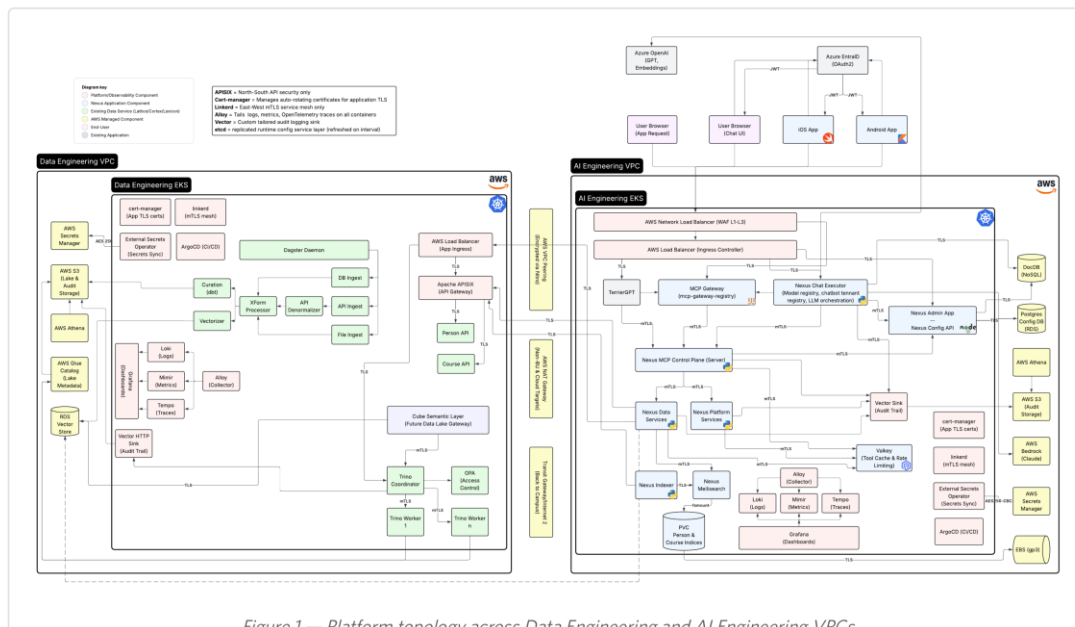


Figure 1 — Platform topology across Data Engineering and AI Engineering VPCs

BU Nexus — Data Engineering VPC (left) and AI Engineering VPC (right), joined only across a gated seam. Source: nexus-platform-overview.pdf p.3.

Seven details the text did not give us

- TWO physically separated VPCs — a Data Engineering VPC (S3 lake, Glue, Athena, Trino, Spark, Dagster, OPA) and an AI Engineering VPC (TerrierGPT, Nexus apps, MCP, pgvector, DocumentDB). They are NOT one network.
- The AI side reaches BU data ONLY through Apache APISIX (API Gateway) across AWS VPC Peering (Nitro-encrypted, TLS), with Linkerd mTLS inside each cluster. Agents never touch the lake directly — exactly the “never reach raw systems” intelligence-stack rule, enforced at the network layer.
- Three egress paths are drawn explicitly: VPC Peering (to BU data), NAT Gateway (non-BU / cloud targets e.g. Azure OpenAI), and Transit Gateway / Internet2 (back to campus systems).

- MCP is the architectural SPINE, not a feature: there is an “MCP Gateway (mcp-gateway-registry)” plus a “Nexus MCP Control Plane.” Tools are registered in a registry and routed by a control plane.
- The concrete data services exposed across the seam today are Person API, Course API and a Cube Semantic Layer (labelled “Future Data Lake Gateway”), with Person & Course vector indices in pgvector. OnBase appears NOWHERE in the topology — a genuine white space.
- Audit is first-class infrastructure: a dedicated Vector HTTP Sink (Audit Trail) on the data side and a Vector Store (Audit Trail) on the AI side, beside the Loki/Mimir/Tempo/Alloy/Grafana observability stack on BOTH VPCs.
- Identity is Azure EntraID (OAuth2); inference is multi-cloud — Azure OpenAI (GPT/embeddings) + AWS Bedrock. OPA (Access Control) gates Trino access on the data side.

Why this matters — OnBase is the missing node

BU's grounding corpus today is Person, Course, WordPress, ServiceNow and student systems. OnBase — the system CogniBase masters — is not yet in the Nexus data plane at all. CogniBase therefore does not displace anything on this diagram; it fills an empty slot. That is the single most encouraging detail in the document: the OnBase node BU will eventually need does not exist, and QI already has it.

Refined integration mechanics for CogniBase (now concrete)

Option	Where CogniBase lands	Mechanism	Verdict
A — AI-side MCP (lead)	AI Engineering VPC	Register CogniBase's MCP server in the mcp-gateway-registry; the Nexus MCP Control Plane routes agent tool-calls to it (search / ask / schema_explain / federated_query).	Lightest, MCP-native, matches the spine. DO THIS FIRST.
B — Data-side API	Data Engineering VPC	Deploy CogniBase as an “OnBase API” service behind APISIX, a peer of Person API / Course API, reachable across the VPC peering; emit curated OnBase entities to the S3/Iceberg lake + pgvector indices.	Heavier but lands OnBase in the durable data plane. PHASE 2.
C — both	Both VPCs	MCP tool surface for agents (A) backed by a governed OnBase data service (B).	End-state. The MCP face is the front door; the data service is the foundation.

What the diagram confirms about the price of entry

- Entra ID (Azure) identity is non-negotiable — the diagram shows it wiring every channel. CogniBase's planned Entra adapter moves from “nice” to “required for option A.”

- mTLS (Linkerd) + APISIX + OPA are the in-cluster contract — CogniBase's Gates map cleanly to OPA; its L1–L4 audit trail matches BU's dedicated audit sinks. The compliance posture is already aligned.
- Per-corpus pgvector indices (Person, Course) are the precedent — CogniBase emitting an “OnBase” pgvector index is the obvious, in-pattern Phase-2 artefact (alongside, not instead of, lake/Iceberg output).

One correction to Rev 1

Rev 1 treated “Iceberg-on-S3” as the universal storage target. The diagram clarifies a split: the Data Engineering VPC holds the S3 lake / Glue / Trino, while the AI Engineering VPC grounds assistants on pgvector indices + DocumentDB (conversations) + Postgres (config). CogniBase's storage-emit recommendation is therefore two-pronged — pgvector index for the AI plane (fast path) and lake/Iceberg for the data plane (durable path) — not a single Iceberg target.

3. The Quiddity Toolset (what we are fitting)

The QI ecosystem holds 25 registered projects. Most are unrelated to BU (consumer/AI apps). The BU-relevant set is small and sharp — three direct-fit products plus a supporting platform layer.

BU-relevant QI products

Project	Path / status	One-liner
CogniBase	C:\CogniBase — pre_poc	Vendor-neutral local OnBase intelligence: schema map + metadata + RAG + federated retrieval, pluggable LLMs (Claude/OpenAI/Gemini/Ollama).
MapSnap	C:\MapSnap — active_stable	Local-first schema-intelligence tool: any enterprise DB → navigable browser + AI that explains schema and does NL→SQL, regulated data never leaves the box.
AutoPDF	C:\AutoPDF — active dev	Local-first PDF toolkit: bulk convert / split / extract → CSV/XLSX/OnBase DIP. No cloud, no telemetry.

Supporting QI platform layer (credibility & integration assets)

Asset	Why it matters to BU
OpenClaw (C:\OC, active_production)	Same platform BU chose for TerrierAI Assistant — direct operational credibility and a concrete collaboration entry point.
QI Brain (MCP server, :9011)	Demonstrates QI is already MCP-native with a knowledge-graph/decision store — mirrors BU's ontology ambition.
QI Hive + 7 sub-agents	Multi-agent orchestration (architect/builder/inspector/ops/scout/scribe/tester) — parallels BU's Regent agent-runtime direction.
Maia (multi-channel AI assistant)	Chat/voice/LINE/Telegram delivery — parallels TerrierAI Agents' chat/voice/SMS/email channel vision.
NEXUS (C:\NEXUS) — NAME COLLISION	Renne's NEXUS collides with BU's Nexus platform name.

	Rename / differentiate before any BU contact.
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4. Alignment Map — QI tools → BU tiers & components

Where each QI tool lands

QI tool	Best BU tier	Plugs into (Atlas)	Integration surface
CogniBase	Tier 01 design partner / Tier 04 tool	Nexus (MCP), Cortex (lake), Lattice (API)	Expose retrieval + Query Federator as MCP tools; emit Iceberg/Trino output to Cortex.
MapSnap	Tier 04 specialized tool	Cortex, Convoy	NL→SQL + schema map as an MCP tool; feed migration discovery for SIS/Jenzabar retirement.
AutoPDF	Tier 04 specialized tool	Convoy (Prefect), OnBase	Batch extraction as a Prefect-runnable job; OnBase DIP output for records ingestion.
OpenClaw expertise	Tier 02 capability injection	Nexus / Assistant	Forward-deployed hardening of the OpenClaw deployment BU is piloting.
Ontology/semantic methodology	Tier 02/03	Lexicon, Regent	Knowledge-graph + Normalizer methodology onto the planned enterprise knowledge graph.

5. Deep Dive — CogniBase (the lead play)

CogniBase is the most strategically important QI asset for BU because it solves a problem BU has publicly and a problem BU's own team has privately. Publicly: BU runs Hyland OnBase and is building AI over institutional data. Privately (per the QI coordination doc): BU has a department exploring the OnBase REST API that has REST/HTTP skills but no deep OnBase domain experience — exactly CogniBase's moat.

What CogniBase already does right for BU

- Vendor-neutral, pluggable LLMs (Claude/OpenAI/Gemini/Ollama) — matches BU's curated-heterogeneity, no-single-bet rule.
- Local-first with a compliance control plane — three-corpus model keeps PII out of admin chat; five-layer access model (Constraint→Normalizer→CrossLink→Gate→Privacy Transformer); k-anonymity floor; L1–L4 audit trail. This exceeds most vended tools and is a FERPA selling point.
- Query Federator solves OnBase's prefix-join problem (AP_Name vs HR_Name) that OnBase Custom Queries cannot — genuine domain IP.
- Already Docker / docker-compose deployable with bind-mounted state — a real head start on K8s.

- “Beyond OnBase” mission (lift OnBase into a vector+lake layer) is the same direction as BU's Cortex lake.

Gap analysis — to be BU-adoptable

Dimension	CogniBase today	BU requires	Action
Deployment	Local-first / Docker on a single host	EKS + Graviton (ARM64), ArgoCD GitOps	Publish an ARM64 image + Kustomize/Helm manifest; verify on arm64.
Agent interface	Internal RAG/agent loop	MCP tool endpoints Nexus/Regent call	Wrap retrieval + federation as an MCP server (nexus-mcp-server-compatible). HIGHEST LEVERAGE.
Storage format	ChromaDB + local schema.json	Iceberg-on-S3, Trino-queryable, pgvector	Add an Iceberg/pgvector emit path so curated OnBase entities land in Cortex.
Identity	CogniBase admin role / AD passthrough	Entra ID (MSAL PKCE), per-user isolation	Implement the already-planned pluggable identity against Entra ID.
Gateway/auth	Local login	APISIX consumer keys, OPA policy	Front with APISIX; express Gates as OPA-checkable policy.
Licensing	Product	No per-seat, freedom to leave	Flat institutional licence + open exit; keep schema/artefacts as plain files (already true).

CogniBase improvement roadmap (to reach BU)

- M1 — MCP wrapper: expose search / federated-query / schema-explain as MCP tools (days, highest impact).
- M2 — ARM64 container + Helm chart; smoke-test on Graviton-class instance.
- M3 — Entra ID identity adapter (MSAL PKCE) replacing local login for BU mode.
- M4 — Iceberg/pgvector emit path feeding Cortex; Trino-queryable curated entities.
- M5 — OPA policy export for Gates; APISIX front door.
- M6 — Joint OnBase test instance with BU REST POC team; vendor-or-share REST adapter decision.

6. Deep Dive — MapSnap (the migration accelerator)

BU's 2026 roadmap includes Jenzabar Retirement, SIS Stabilization, mainframe archive, SAP S/4 evaluation, and dozens of source-system ingestions into the data lake. Every one of those is a schema-archaeology problem — understanding an undocumented, FK-less enterprise schema before you can migrate or ingest it. That is precisely what MapSnap does, and it already supports SQL Server, Oracle, Postgres, MySQL, SQLite, OnBase, and Jenzabar EX.

Fit highlights

- NL→SQL over a 2,000-table schema with relationships inferred (the prior OnBase run produced 34 MRG-derived + 6,167 inferred soft FKs) — turns months of reverse-engineering into minutes.
- Local-first egress guardrail (local_only by default) — regulated data never leaves the box; directly answers BU's FERPA/GLBA concern about “chat with your database” tools.
- Engine-agnostic sidecar knowledge bundle (annotations, value catalog, profiling, vector index) — the same enrichment chain that becomes CogniBase; a coherent product family story.

Gap analysis

Dimension	MapSnap today	BU requires	Action
Packaging	Windows, Python HTTP server (:9876), NSSM, Cloudflare tunnel	Container/K8s or analyst laptop tool	Containerize, or position as a Tier-04 analyst desktop tool (lighter lift).
Agent surface	Standalone HTML app	MCP tool for Nexus agents	Expose schema-explain + NL→SQL as MCP tools Regent/Nexus can call during migration work.
Local LLM	Ollama only	Curated heterogeneity, cloud-gated	Use existing policy-gated AI Connections to surface Claude via Bedrock, matching BU.

Best framing for BU

Position MapSnap not as a product to buy but as the discovery layer of a migration engagement: “point it at Jenzabar EX / the SIS / the mainframe extract and get a plain-English, join-aware map in an afternoon.” Pair with BU's Convoy/Prefect + Claude Code conversion skill — MapSnap finds and explains; Convoy converts.

7. Supporting — AutoPDF (the records on-ramp)

AutoPDF is the third OnBase-aligned product: a local-first toolkit that bulk-converts, splits, and extracts metadata from document piles and emits a schema-accurate OnBase XML Index DIP (Hyland Foundation 24.1). For BU's records / imaging / registrar / procurement teams it is the ingestion on-ramp that feeds OnBase — with confidential documents never leaving the machine.

Fit & gap (brief)

- Fit: local-only, no telemetry, rule-based-first with optional local-Ollama assist — FERPA-safe; OnBase DIP output is exactly what records teams need.
- Gap: bundled GPL/AGPL engines (Ghostscript/Poppler/PDFtk/NAPS2) are kept compliant via arm's-length separate-process aggregation — document this clearly for BU's open-source governance (Apache-2.0/permissive-preferred, no-GPL-in-core-infra policy). As a standalone desktop tool the GPL aggregation is fine; do NOT embed these libraries into any BU core service.
- Best framing: a Tier-04 desktop utility for records teams during the OnBase content build-out; bundle with CogniBase (ingest → understand → retrieve).

8. Strategic Overlaps, Collisions & Convergence

OpenClaw — the credibility bridge

BU's TerrierAI Assistant is built on OpenClaw, currently in “deployment hardening / pilot scoping.” Renne runs OpenClaw in production today. This is the single most under-used asset: it converts QI from “outside vendor” to “practitioner who has already operated the platform BU is adopting.” Offer OpenClaw operational patterns, MCP tools, and hardening notes as a no-cost relationship opener.

MCP + Claude Code — the technical handshake

BU's agents reach data exclusively through MCP tools, and BU uses a custom Claude Code skill for SnapLogic→Prefect conversion. QI is already MCP-native (QI Brain) and Claude-Code-operated end to end. Any QI tool exposed as an MCP server is natively callable by Nexus/Regent — making MCP wrappers the highest-leverage engineering investment across the whole portfolio.

The NEXUS naming collision — fix before contact

BU's flagship AI platform is “Nexus.” Renne's own project C:\NEXUS shares the name. In any BU-facing document, deck, or repo this will read as either confusion or presumption. Recommendation: rename Renne's NEXUS (or give it a clear product name) for external materials, and never present a QI “NEXUS” to BU.

Differentiation vs. CollegeVine

CollegeVine occupies the Tier-01 design-partner slot but is admissions/recruiting-focused. The operational / IT-data / records / migration domain is wide open. QI should position there — agents and tools for administrative and data-engineering workflows, not student-facing recruiting.

9. History of Fitting QI Work into BU — and what it implies

QI has already invested in the BU relationship along three lines, captured in on-disk artefacts:

What has happened so far

- CogniBase was conceived specifically as “CogniBase @ BU” — the OnBase descendant of MapSnap, with a full design set (architecture, access model, query federator, hygiene pipeline) and a dedicated BU-REST-POC coordination document.
- The coordination doc establishes the relationship model: “coordinate, don't integrate” — both teams hit the same Hyland REST API; QI owns the OnBase domain layer, BU owns departmental DocType inventory and credentials. Several open items remain (BU contact, joint test instance, credential issuance).
- BU IT/licensing documents and explainer videos (Andrew/Ava) were produced for MapSnap and AutoPDF (2026-06-27) — IT-facing component + open-source-licensing briefs, ready to hand to BU IT.
- An inverted, technology-first ecosystem tech-stack reference exists for all QI apps — useful for demonstrating open-standard alignment to BU's governance team.

What the history implies

- The intellectual work is done; the missing pieces are relational and packaging, not conceptual: a named BU contact, issued credentials, a joint test instance, and MCP/container packaging.
- QI should convert the coordination doc's open placeholders into a concrete ask: one IT request for credentials usable by both teams against a shared OnBase test instance.
- The “Consulting Partner AI Landscape/Roadmap Review” line on BU's roadmap is a named, RFP-shaped opening — QI's tech-stack reference + these BU IT docs are exactly the credibility package for it.

10. Adoption Price-of-Entry — per-tool conformance scorecard

Scored against BU's six stated architecture rules. ✓ = already meets, ① = partial / planned, X = gap to close.

Conformance

BU rule	CogniBase	MapSnap	AutoPDF
Open-standard native (MCP/Iceberg/Trino/OPA)	① (RAG yes; MCP/Iceberg to add)	① (NL→SQL; MCP to add)	✓ (open formats, DIP/CSV/XLSX)
No per-seat / freedom to leave	✓ (plain-file artefacts)	✓	✓ (copy-a-folder)
Identity-aware & auditable (Entra ID)	① (audit ✓; Entra to add)	① (roles ✓; Entra to add)	X (local PIN/tunnel)
Data-classification guardrails (FERPA)	✓ (Gates + Privacy Transformer)	✓ (egress guardrail)	✓ (local-only)
Cloud-native / ARM / K8s	① (Docker ✓; ARM/Helm to add)	X (Windows/NSSM)	X (Windows desktop)
Forward-deployed delivery model	✓ (engagement-ready)	✓	✓

Read of the scorecard

CogniBase is closest to BU-ready and carries the deepest moat — it should lead. MapSnap and AutoPDF are already compliant on the things that are hard to retrofit (open formats, FERPA posture, no-lock-in) and fall short only on cloud/identity packaging, which is acceptable if they are positioned as Tier-04 analyst/records desktop tools rather than core-platform services.

11. Commercial & Pricing Directions

BU's stated stance — “AI has changed build vs. buy; we build in-house” — makes a pure product sale a hard sell. The winning commercial shapes are co-development, capability injection, and flat-fee tools that respect “freedom to leave.” Per-seat licensing is structurally disqualifying. The budget context is favourable: BU is actively eliminating ~\$450K/yr (SnapLogic) and re-licensing MicroStrategy — freed money that an open, migration-helping tool can partly capture.

Three commercial models (use in combination)

Model	Shape	Indicative pricing*	Best for
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Design partnership (Tier 01)	Forward-deployed co-development; QI embeds and builds the OnBase domain layer BU lacks. Reference + roadmap influence in exchange for low upfront.	Low/no licence upfront; \$8–15K/mo forward-deployed engagement, or revenue-share / milestone SOW	CogniBase + the REST POC team
Specialized tool licence (Tier 04)	Flat institutional licence, open exit, source/artefacts as plain files. No per-seat.	\$15–40K/yr per tool flat, + optional support tier	MapSnap, AutoPDF, CogniBase-as-tool
Consulting / capability injection	Respond to BU's named "Consulting Partner AI Landscape/Roadmap Review"; scoped advisory + accelerators.	\$150–250/hr or fixed-scope \$20–60K project	Roadmap review, ontology/semantic methodology

Pricing principles

- Anchor against what BU is removing, not against commercial SaaS list prices — “a fraction of the \$450K/yr you are freeing” is the right framing.
- Never quote per-seat. Quote per-tool-flat, per-engagement, or per-outcome.
- Always include an open-exit clause and plain-file data portability — it turns BU's “freedom to leave” test from a risk into a selling point.
- Lead with a low-commitment pilot (joint OnBase test instance) before any licence — BU's culture is pilot-then-scale.
- * All figures are planning placeholders for Renne to set against actual cost and appetite — not quotes.

12. Recommended Directions (ranked)

Ranked plays

#	Direction	Why now	Effort	Impact
1	CogniBase as the OnBase domain layer for BU's REST POC team	Lowest friction — they have REST plumbing, lack OnBase domain depth; CogniBase is that exact layer.	Medium	Very High
2	Bundle MapSnap + AutoPDF as a Jenzabar/OnBase migration toolkit	BU is retiring Jenzabar & stabilising SIS now; schema archaeology + records ingestion are immediate needs.	Low–Medium	High
3	Wrap CogniBase & MapSnap as MCP servers	Makes them natively callable by BU's Nexus/Regent agents; smallest change with the	Low	Very High

		largest compatibility payoff.		
4	Respond to the "Consulting Partner AI Landscape/Roadmap Review"	A named, RFP-shaped door; QI already has the credibility package (tech-stack ref + BU IT docs).	Low	Medium–High
5	Pursue the Tier-01 design-partner slot in the operational/IT-data niche	CollegeVine occupies admissions; the operational domain is open.	High	High
6	Offer OpenClaw operational hardening as a relationship opener	BU is hardening the exact platform Renne already runs; near-zero cost, high trust-building.	Low	Medium

13. 30 / 60 / 90-Day Action Plan

Next 30 days — packaging & positioning

- Build an MCP wrapper for CogniBase retrieval + Query Federator (and a thin one for MapSnap NL→SQL).
- Rename / differentiate Renne's NEXUS for all external materials.
- Assemble the BU outreach pack: 1-page CogniBase-for-BU brief + the existing MapSnap/AutoPDF IT & licensing docs + tech-stack reference.
- Draft the single IT credential request (shared OnBase test instance for both QI and the BU REST team).

Next 60 days — pilot enablement

- Publish ARM64 container + Helm chart for CogniBase; smoke-test on a Graviton-class instance.
- Implement Entra ID (MSAL PKCE) identity adapter for CogniBase BU mode.
- Stand up a joint OnBase test instance with the BU REST POC team; fill the coordination doc's open placeholders.
- Identify a sponsoring BU team / use case (records, registrar, or data-engineering migration).

Next 90 days — demonstrate value in BU's own terms

- Demo CogniBase answering a real cross-DocType question via the Query Federator on the test instance.
- Demo MapSnap mapping a Jenzabar/SIS schema in an afternoon, output feeding a Convoy/Prefect conversion.
- Add an Iceberg/pgvector emit path so curated OnBase entities land in Cortex (TerrierData Lake).
- Propose a scoped engagement: design partnership SOW or flat-fee migration-toolkit licence.

14. Risks & Mitigations

Key risks

Risk	Mitigation
"We build in-house" → product sale rejected	Lead with co-development / forward-deployed capability injection, not a product pitch.
"Freedom to leave" → lock-in resistance	Plain-file artefacts, open formats, open-exit clause as a feature.
Local-first vs. cloud-native mismatch	Ship ARM64 containers + Helm; or position Windows tools explicitly as Tier-04 desktop utilities.
Looking like the middleware BU is removing (SnapLogic/MicroStrategy)	Emphasise open standards, MCP, no-lock-in; frame as migration *accelerator*, not new middleware.
CollegeVine incumbency in Tier 01	Differentiate into operational/IT-data; do not compete on admissions.
FERPA / data classification	CogniBase Gates + Privacy Transformer; MapSnap/AutoPDF local-only; never ingest HIPAA/PCI.
NEXUS name collision	Rename / differentiate before any BU-facing material.
Relationship stalls (no contact/credentials)	Convert coordination-doc placeholders into one concrete IT ask; use the roadmap review as a formal entry.

15. Appendix — Source Index

Documents & registries read

- C:\Users\renne\Downloads\BU-AI-Data-Engineering-Dossier-for-Quiddity.md
- C:\Users\renne\Downloads\nexus-platform-overview.pdf (8 pp)
- C:\QIH\ecosystem\qi_registry.json (25 projects)
- C:\CogniBase\README.md, DESIGN\ARCHITECTURE.md, COORDINATION_BU_TEAM.md, STRATEGY_BEYOND_ONBASE.md
- C:\MapSnap\INTRO\status_intro.md + docs
- C:\AutoPDF\INTRO
- Memory: project_bu_techdocs_videos.md (BU IT/licensing docs + explainer videos)